



UNIVERSITY OF LEEDS

Formal Language & Finite Automata

Variants of Turing machines

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All reading is from the "Introduction to the theory of computation" - Micheal Sipser

- Turing machines - Pages 176 - 184 (essential)

Multitape Turing machines

Definition of multitape Turing machine

- A multitape Turing machine M is a TM with multiple tapes each with their own head.
- A multitape Turing machine starts with each head on the first cell and the first tape containing the input while all other tapes being empty.
- For a multitape Turing machine with k tapes the transition function has the form:

$$\delta : Q \times \Gamma^k \rightarrow Q \times \Gamma^k \times \{L, R, S\}^k.$$

- Here $\delta(p, a_1, \dots, a_k) = (q, b_1, \dots, b_k, D_1, \dots, D_k)$ means that if M is in state p and the head of its i -th tape reads a_i for every $i \in \{1, \dots, k\}$ then it transitions into state q , writes b_i onto its i -th tape for every $i \in \{1, \dots, k\}$ and moves the head of the i -th tape left if $D_i = L$, right if $D_i = R$ and stays in its current position if $D_i = S$ for every $i \in \{1, \dots, k\}$.

Equivalence of TM's and multitape TM's

Theorem

Every multitape Turing machine has an equivalent single tape Turing machine.

Corollary

A language is Turing recognizable/decidable if and only if some multitape Turing machine which recognizes/decides it.

Non-deterministic Turing machines

Definition of non-deterministic Turing machine

- Non-deterministic Turing machines are defined in the same way as TM's only the transition function is of the form

$$\delta : Q \times \Gamma \rightarrow \mathcal{P}(Q \times \Gamma \times \{L, R\}).$$

- Computation of a non-deterministic TM is a tree whose branches correspond to all different possibilities of how the computation proceeds.
- A non-deterministic TM accepts a word w if at least one branch of computation leads to an accepting state.
- A non-deterministic TM rejects a word w if all possible branches of computation reject w .

Equivalence of non-terministic and determinic TM's

Theorem

Every non-deterministic Turing machine has an equivalent deterministic Turing machine.

Corollary

A language is Turing recognizable/decidable if and only if there is a non-deterministic Turing machine which recognizes/decides it.

Algorithm

Algorithm vs. Turing machine

Informal: Every sensible model of computation that allows arbitrary access to memory and each step of computation performs only a finite step of work will be equivalent to a Turing machine.

Informal: Everything an algorithm can compute can be computed by a Turing machine (Church-Turing thesis says that λ -calculus is equivalent to Turing machines).

Consequence: The formalization of the term algorithm allows us to prove that there are problems for which there is no algorithm (they are not Turing decidable).